MPFM — CANONICAL FIXED CORE

1. Ontological Status of Information

Fixed Principle

> Information is fundamental. Dynamics is secondary.

- Information exists prior to space, time, matter, and energy.
- Physical reality emerges from non-faithful projections of information.
- No physical observable exhausts total information.

2. Projection Principle

Fixed Principle

> Every physical observation or interaction is a projection that irreversibly loses information.

Formally:

$$\Delta \mathcal I \;>\; 0 \quad \text{for every projection}$$

- Projections are non-invertible.
- Lost information is not hidden, not recoverable, and not encoded elsewhere.

3. Information Categories (Fixed Classification)

Category	Status
Pre-projective information	Non-observable, non-measurable
Projected information	Context-dependent observables
Lost information	Universal, irreversible
Robust information	Invariants under projection
Organized information	Stable attractors (matter, life, mind)

4. Information Loss Measured in Bits

Fixed Definition

> A bit is a unit of irreversible loss of distinguishable possibilities.

- Bits are measures, not symbols.
- Information loss is quantified logarithmically.
- Bits unify entropy, decoherence, and gravitational entropy.

5. Hierarchy of Minimal Information Loss ($\Delta\mathcal{I}$)

Fixed Hierarchy

$$\Delta I_{\text{pre}} < \Delta I_{\gamma} < \Delta I_{f} < \Delta I_{\text{atom}} < \Delta I_{\text{macro}} < \Delta I_{\text{life}} < \Delta I_{\text{cons}} < \Delta I_{\text{horizon}}$$

Meaning:

- Different structures have different irreducible loss floors.
- No universal minimal bit exists.

6. Spectral Dimension ($D_s < 3$)

Fixed Result

> The effective spectral dimension of projected information is non-integer and strictly smaller than 3.

- $D_s < 3$ explains:
  - turbulence regularity,
  - decoherence scaling,
  - residual noise,
  - cosmological correlations.

7. Law of Residual Informational Noise

Fixed Law

$$\boxed{\Delta I(L) \;\propto\; L^{2-D_s}}$$

- Non-exponential
- Non-integer power law
- Universal across scales

8. Experimental Priority (Fixed)

Single Priority Experiment

> Measurement of extragalactic radio polarization correlations versus comoving distance (SKA / LOFAR / JVLA).

Prediction:

$$-\ln C\,(L) \sim L^{2-D_s}$$

This test is:

- falsifiable,
- adjustment-free,
- unique to MPFM.

9. Causality Principle

Fixed Principle

> Causality is bounded by irreversible information loss, not only by signal speed.

Formally:

$$\text{Causal accessibility} \;\leq\; e^{-\Delta \mathcal{I}}$$

- Speed c is necessary but not sufficient.
- Intrication does not transmit causal information.

10. Stability of Physical Laws

Fixed Explanation

> Physical laws are stable because they describe information invariants that survive projection.

- Laws are not chains of causes.
- Laws are relations between robust attractors.
- Fragile laws cannot exist.

11. Time and the Arrow of Time

Fixed Principle

> Time emerges from the accumulation of irreversible information loss.

- No loss → no time.
- Memory = stabilized projection.
- Future is open because information is not yet lost.

12. Gravity (Fixed Interpretation)

Fixed Definition

> Spacetime curvature is not a property of space itself, but the geometric language used to describe how projections reorganize around what cannot be fully seen.

- Gravity = structural information deficit
- No gravitational charge
- Always attractive
- Not quantizable as a local interaction

13. Einstein Equation (Reinterpreted)

Fixed Interpretation

> Einstein’s equation is an information-balancing equation.

Geometry adjusts to minimize projectable informational deficit.

14. Black Holes

Fixed Statements

- A black hole is a maximal projective boundary.
- Its mass is carried by its boundary.
- Entropy = maximal projectable information loss.
- Evaporation = projection leakage.
- No singularity is physically real.

15. No Exact Prediction Principle

Fixed Law

> No physical theory can make exact predictions beyond what survives projection.

- Even ideal quantum mechanics saturates this bound.
- Perfect laws ≠ perfect predictions.

16. Matter, Charges, and Fermions

Fixed Points

- Charges arise from oriented phase projection (S^1).
- Exactly two signs exist.
- Fermionic statistics emerge from non-differentiability.
- Pauli principle is structural, not imposed.

17. Electromagnetism vs Gravity

Fixed Comparison

EM	Gravity
Phase-based	Phase-less
Signed	Unsigned
Quantizable	Structural
Screenable	Non-screenable

18. Life and Consciousness

Fixed Results

- Life = information attractor that locally slows loss.
- Replication is required by irreversible loss.
- Consciousness = reflexive attractor.
- Consciousness is not simulable by local dynamics alone.

19. Resistance Index ($\mathcal{R}$)

Fixed Definition

$$\mathcal{R} = e^{-\Delta \mathcal{I}}$$

- Quantifies robustness of structures.
- Orders particles → atoms → cells → minds → horizons.

20. What MPFM Is NOT (Fixed Disclaimer)

- Not a gadget theory
- Not a digital universe
- Not numerology
- Not a hidden-variable theory
- Not a TOE promising exact constants

21. Methodological Rule (Fixed)

> MPFM provides constraints, bounds, and forbidden regions — not exact numerical constants.

Any future exact calculation belongs to a V2 independent model.

22. Status of Constants (Fixed)

Constant	Status
c	Structural bound
Λ	Canonical informational scale
a_i	Derived informational acceleration
α	Not calculable (bounded only)
G	Structural normalization

23. Scientific Ethos

> MPFM must be abandoned if a clean experiment refutes it.
> No reinterpretation allowed.

Final Canonical Sentence

The universe is not governed by perfect causality. It is governed by what survives irreversible projection.